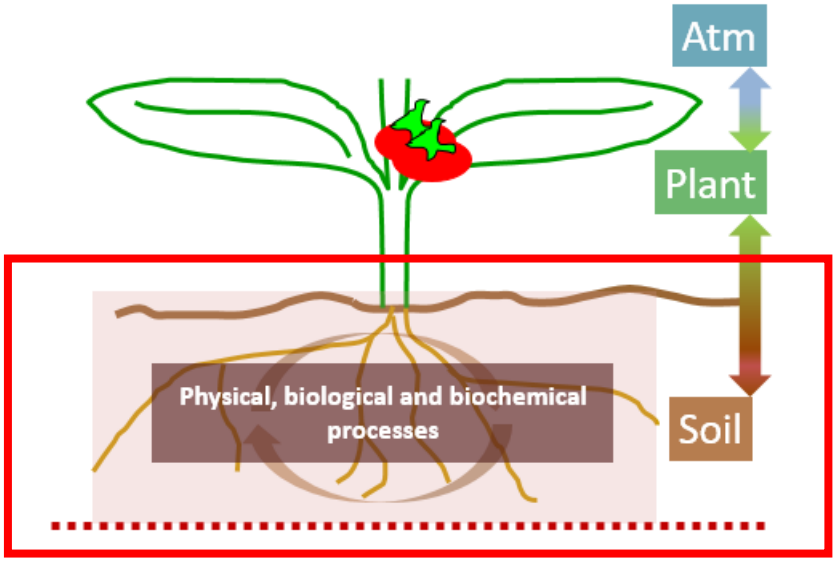
Soil functioning

Outline



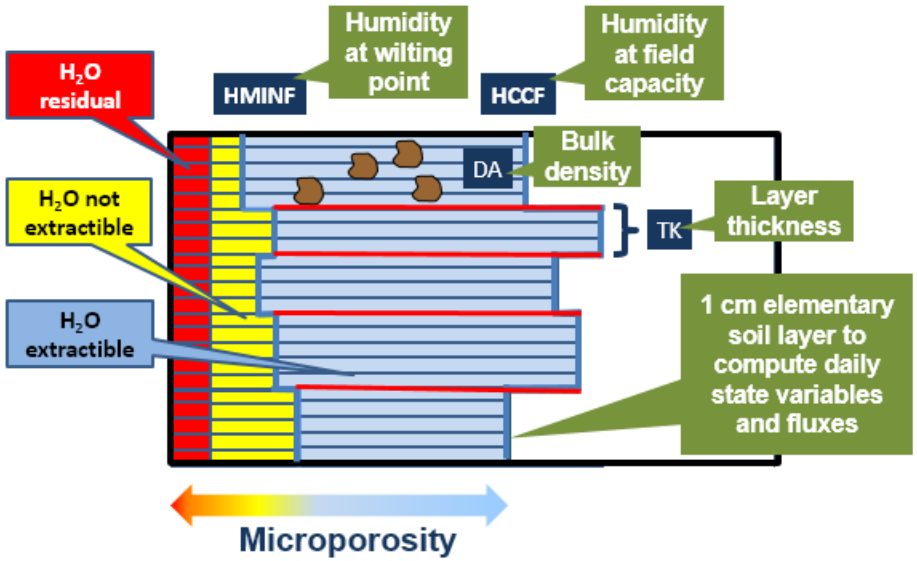
- **Soil compartments and parameters**
- **Boundary conditions, initial conditions**
- **Transfers (water, nitrate, heat)**
- **Soil biogeochemistry (C and N transformations, fluxes)**

Soil functionning

1. Soil compartments and parameters

Soil layers

- **Up to five layers** (~soil horizons) described in the soil input file
- **Surface parameters** (clay content, pH, N content, CaCO3 + albedo, q0, ruisolnu…)
- **Layer specific parameters** : DAF, hccf, hminf, cailloux, epd…



- **Elementary layers** for computational purpose : 1 cm
- **Mixing cell thickness** for nitrat dispersion (epd)

	epc	HCCF	HMINF	DAF	cailloux	typecailloux	infil	epd
layer 1	10	16	10.3	1.53	0	1	50	10
layer 2	10	14	7	1.74	0	1	50	10
layer 3	25	14	6	1.74	0	1	50	10
layer 4	30	13.1	5.3	1.67	0	1	50	10
layer 5	0	0	0	0	0	0	0	0

- **« active » layers** : humification, denitrification
  - do not coincide necessarily with a soil layer
  - specific parameters (denitrification potential…)
- **General parameters** (zesx, cfes, obstarac…)

Soil compartments

Microporosity is mandatory, other compartment are optional

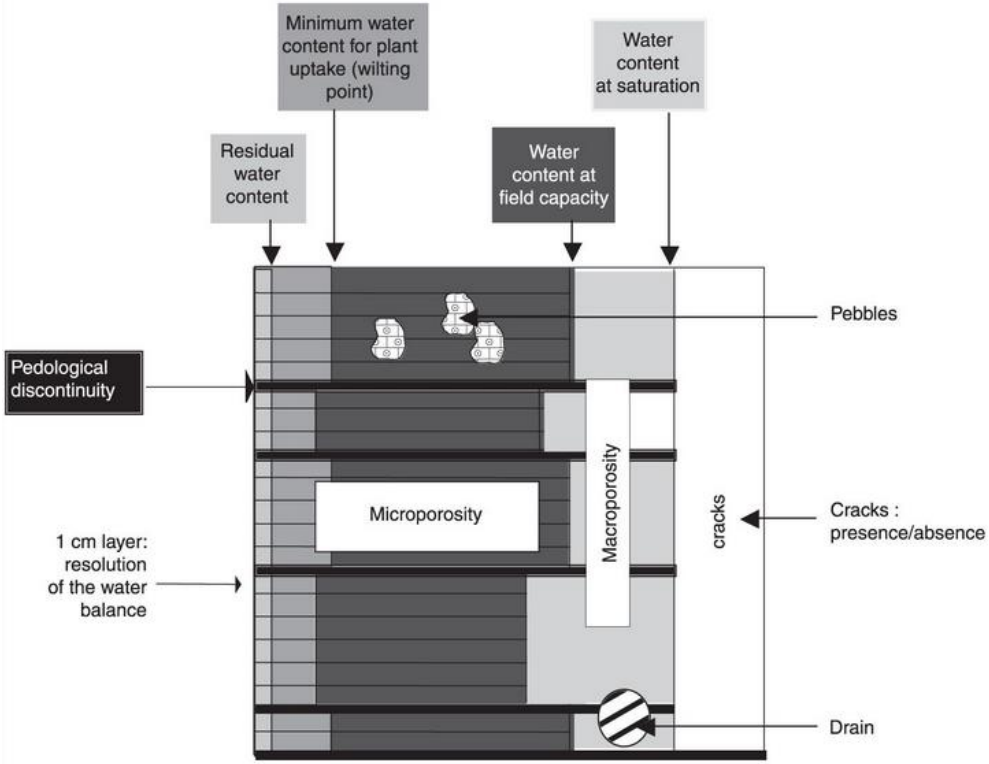
1. **Macroporosity** (codemacropor = 1)
= saturation – field capacity
active if infiltrability at the base of soil layers is limiting, control anoxia, useful to describe the effect of impedding layers (plow pan for ex)

2. **Cracks** (codefente = 1 & codemacropor = 1) for swelling soils

3. **Pebbles** (codecailloux = 1)

4. **Artificial drains** (codedrainage = 1 & codemacropor = 1)

C and N pools (see later)



Soil functionning

2. Boundary and initial conditions

Boundary conditions

- **Soil surface**
  - Water **input** through rainfall (- runoff), irrigation or **losses** through evaporation/transpiration
  - Air temperature (no heat flow calculation)
  - C and N **inputs** (mineral or organic fertilizers, residue, rainfall/irrigation…) or **outputs** (gaseous losses : NH3, N2O, (N2), N uptake

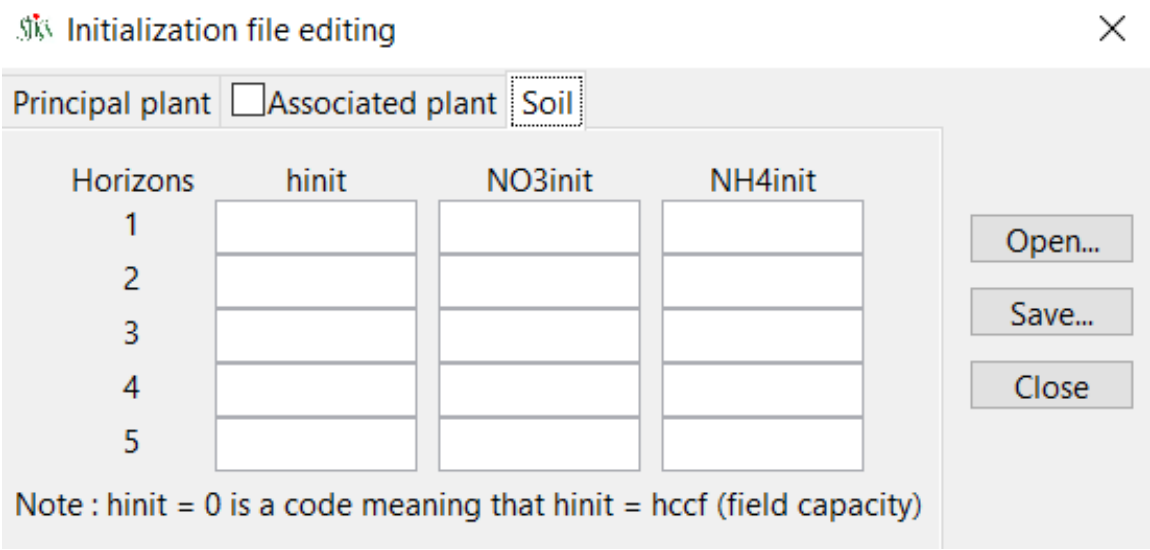
- **Bottom of soil profile**
  - Water **losses** through drainage, or **input** through capillary rise
  - N **losses** through nitrate leaching

- **Modifications due to snow** : preprocessing of climate data
  - Water is stored when snow accumulates and released later at melting
  - Isolation effect of the snow cover : effect on surface temperature

Initial conditions

- **Soil water**: % gravimetric (!) for each layer; 0 means hccf

- **Soil mineral nitrogen** : NH4, NO3, kg/ha



Choosing the best moment for initialization: no plant, known soil water content, mineral N profile available…

Soil functionning

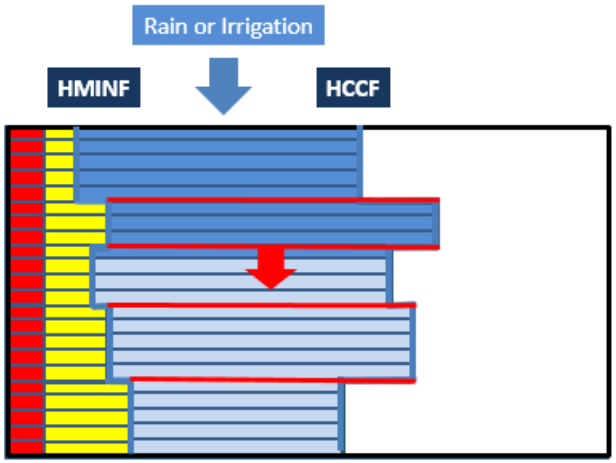
3. Transfers (water, nitrate, heat)

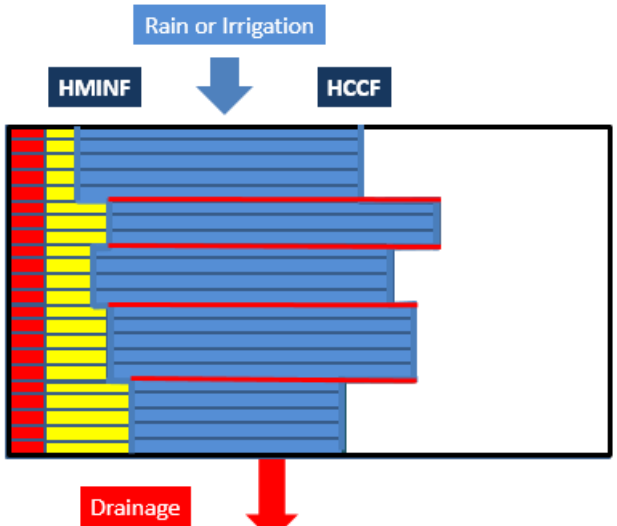
Water transfer

Non preferential downward flow

- Reservoir analogy (reservoir = hccf - hminf)
- Water entering the soil fills the elementary layers from top to bottom, assuming that the maximum soil water content of each layer is its field capacity (flow rate is not limited).
- Drainage can occur when the whole profile is at field capacity

- Soil water can only be removed from each elementary layer through evaporation and plant transpiration



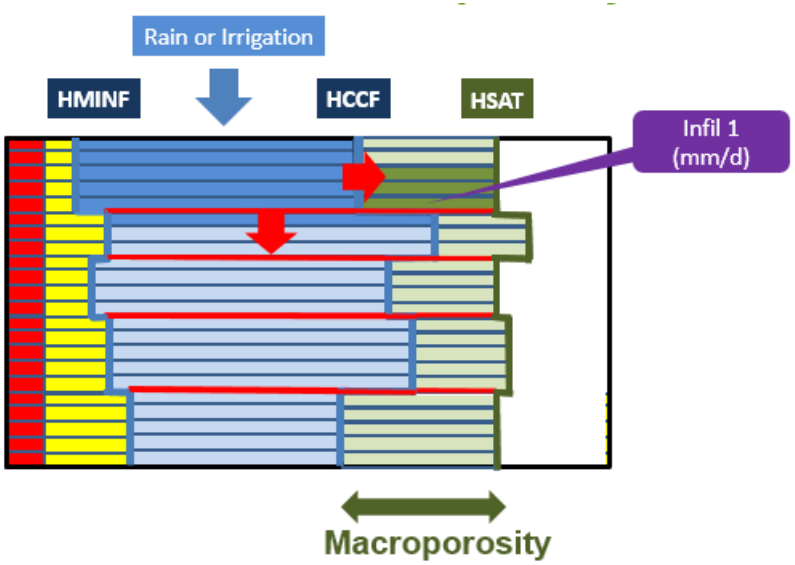


Water transfer

Macroporosity

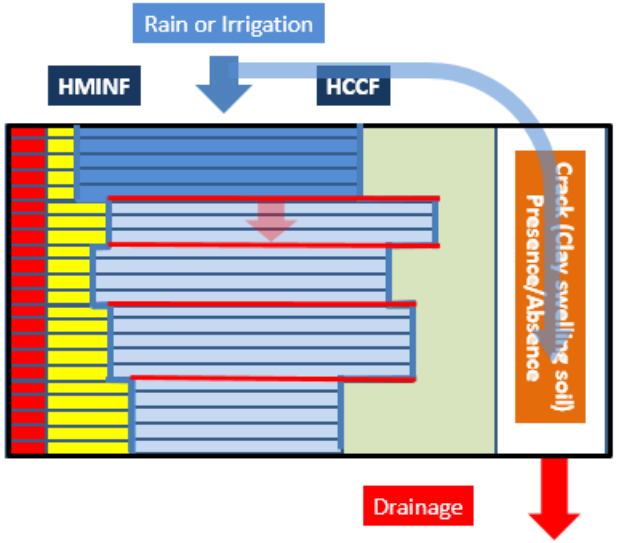
- Limiting infiltrability value at the base of soil layers (one or more)
- When the deepest elementary layer of a soil layer is at field capacity, part of the excess water flow downward (at a rate equal to infiltrability), the remaining flows into macroporosity

-> progressive saturation from the bottom, anoxia



Cracks

- unique compartment, extending from the soil surface to the maximum soil depth, which replaces the macroporosity
- Only feeded from the surface
- Direct pathway to drainage



Water transfer

WFPS option for nitrification and denitrification :

allows to take into account soil water content > field capacity and < saturation for the soil surface for days with rainfall

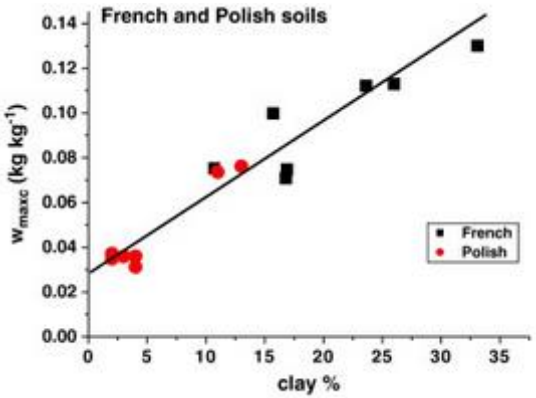
- Rainfall saturates surface layers by filling the empty porosity until saturation
- Progressive desaturation during the day until reaching soil water content at field capacity
- Used for WFPS effect on N2O production/reduction processes
- No output, only internal (does not affect transfers) !

Water transfer

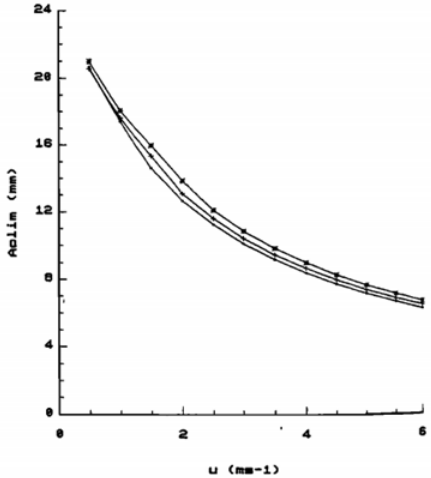
Upward flow: evaporation

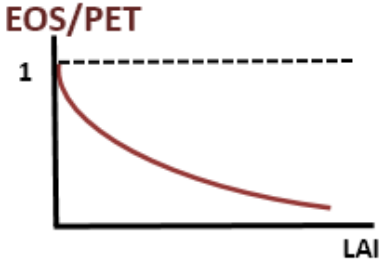
- Potential evaporation depends on climatic demand and vegetative cover
- Actual evaporation = 2 steps

Better HA estimate:



ACLIM estimate:



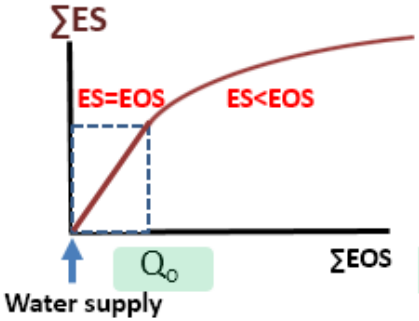


Potential Soil Evaporation

$$EOS(I) = TETP(I)\, exp(-DELTA \cdot LAI(I))$$

Reference potential evapotranspiration

F(crop)



Actual Soil Evaporation

$$\sum_{\substack{beginning \\ second\ phase}} ESOL\ (I) = \sqrt{2A \sum_{\substack{beginning \\ second\ phase}} EOS\ (I) + A^2} - A$$

and

$$A = \frac{1}{2} ACLIM_C (0.63 - HA)^{\frac{5}{3}} (HX_S - HA) \quad with \quad HA = \frac{ARGI_S}{1500}$$

F(weather)

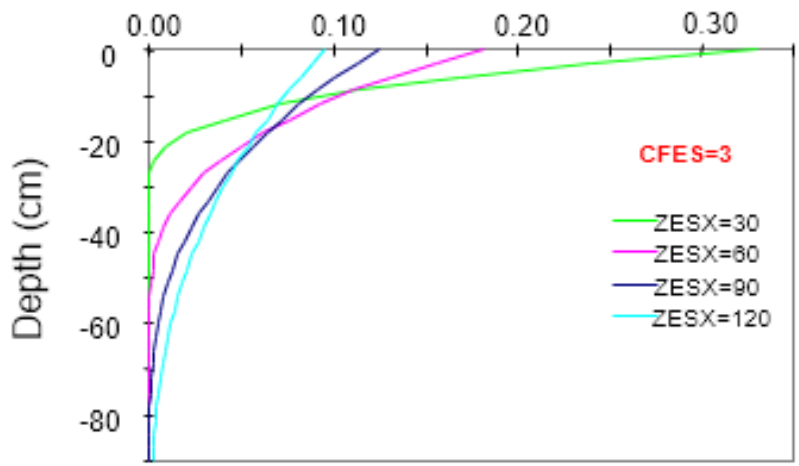
Hum FC

% Clay of soil

Water transfer

Evaporation profile

- Maximum depth
- Shape

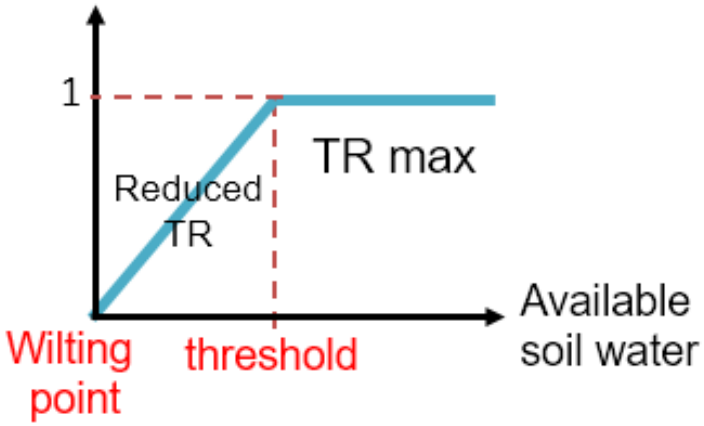


zesx : maximum depth contributing to the evaporation
CFES : parameter describing the shape of contribution of the soil by evaporati

Soil also influence transpiration :

- Evaporation has priority on transpiration
- Threshold of available water below which transpiration is reduced when the soil is drying

EP/EOP = actual/potential transpiration



Nitrate transfer

- Nitrate flows with water (convection)
- Any nitrate arriving into an elementary layer mixes with the nitrate already present
- Excess water flows downward with the new concentration of the mixture
- Dispersion is taken into account through « mixing cells » which thickness (epd) is higher than elementary layer thickness

Table 6. Dispersivity (predicted using the regression coefficients in Table 5) as a function of soil textural class.†

Textural Class	n	ψ_a	b	α_{pred}
		kPa		10^{-3} m
Sand	14	0.69	2.79	8
Loamy sand	30	0.36	4.26	26
Sandy loam	124	1.41	4.74	34
Loam	103	3.55	5.25	46
Silt loam	394	7.59	5.33	56
Sandy clay loam	104	1.35	6.77	60
Clay loam	147	2.63	8.17	81
Silty clay loam	324	6.17	8.72	96
Sandy clay	16	0.98	10.73	109
Silty clay	43	3.24	10.39	110
Clay	148	4.68	11.55	128

† Mean values of ψ_a and b from Cosby et al. (1984).

epd = 2 * α

Perfect et al. 2002

Heat transfer

- No heat flow calculation!
- Temperature at the soil surface is equal to tcult : determine the daily thermal amplitude
- Tsol at depth z calculated by propagating the surface wave taking into account the exponential attenuation of the amplitude due to inertia

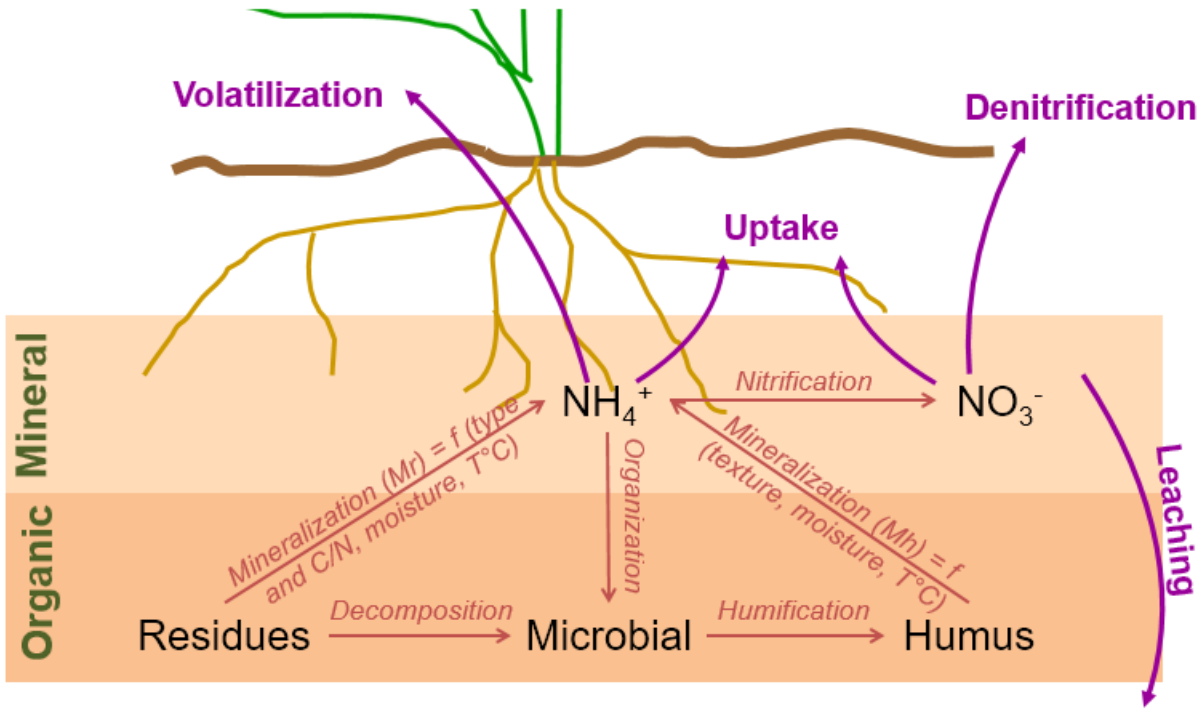
Soil functionning

4. Soil biogeochemistry (C and N transformations, fluxes)

C N transformations

- Different pools that are related
- Active layers (prohhum, profdenit)
- Interaction with transfers (local conditions, transport)

- Potential rate + modulation by substrate availability & environmental conditions (temperature, water content, aeration, pH…)



Pools and fluxes

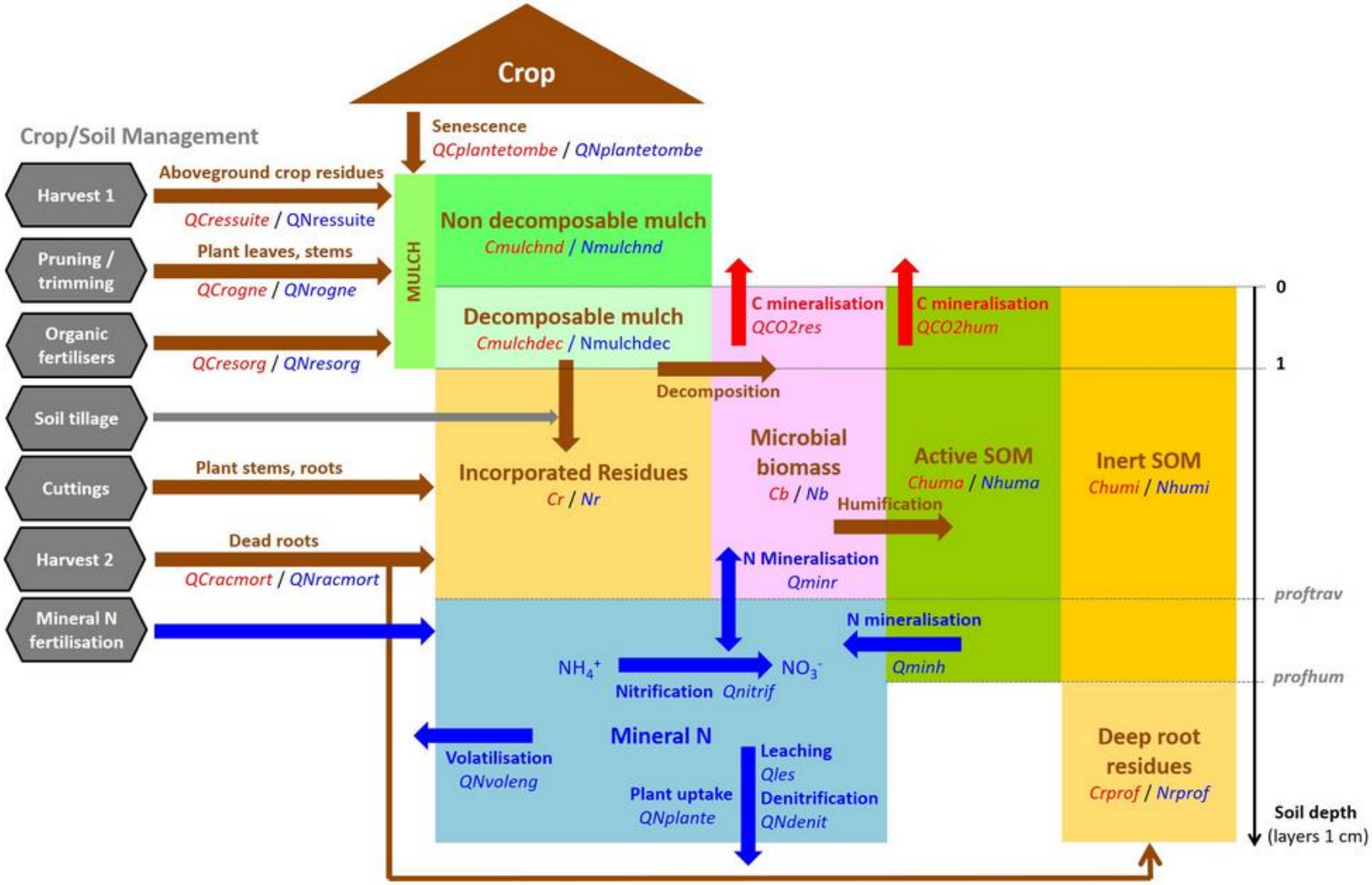


Figure 12.1. Soil C and N compartments and incoming and outgoing C and N fluxes in the STICS model (adapted from Autret *et al.* (2020)). Concerning arrows and text, grey is used for management practices, brown for biomass / organic matter, red for carbon and blue for nitrogen. STICS variable names are in italics.

Overview

Decomposition/mineralization

- 3 organic pools : residue, microbial biomass, humus (already present or newly formed)
- rate dependent on C:N ratio, soil water content, temperature
- induces mineralisation or organisation, which influences the mineral nitrogen pools
- Humus also mineralized

Nitrification

- transforms NH4 into NO3
- part of the nitrified N is lost as N2O

Mineral nitrogen (NH4 or NO3) can be used by the plant (without selectivity)

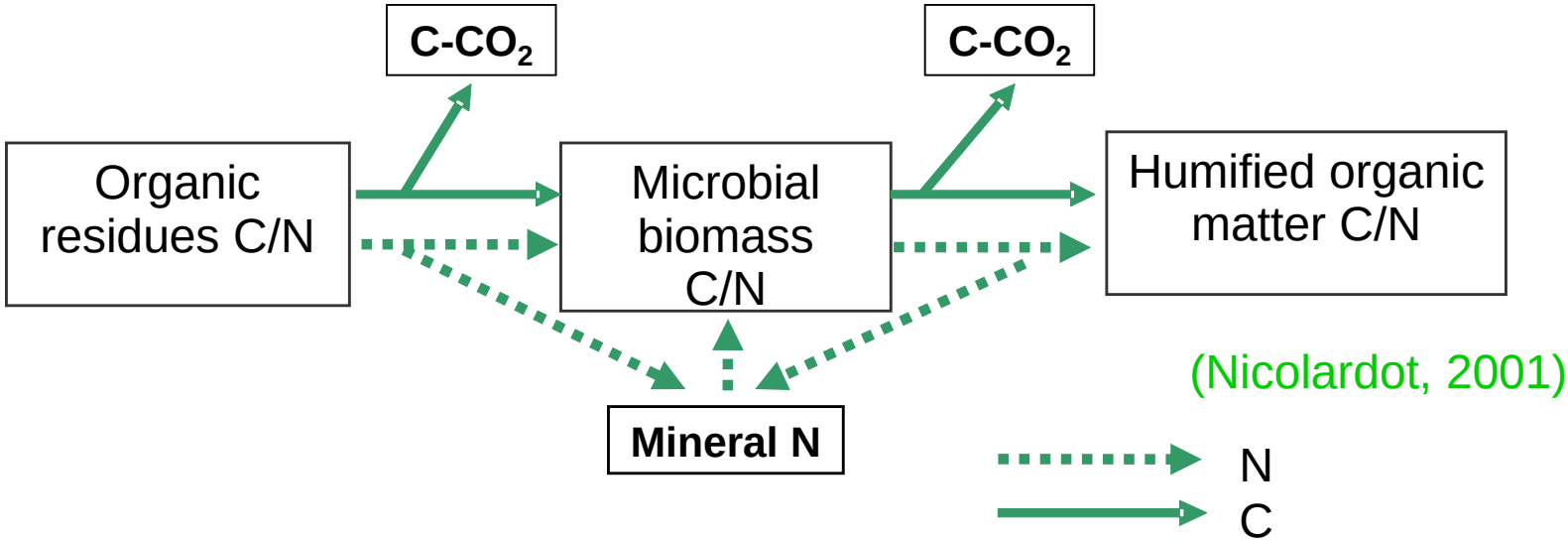
Denitrification

- NO3 is transformed in N2O
- N2O can be reduced to N2

No N2 pool in STICS (gaseous loss which can be estimated from the N balance)

Residues decomposition and mineralization

Decomposition = f(Soil moisture and temperature (Normalized days), residues nature and C/N, available SMN)



(Nicolardot, 2001)

Successive regulation according to N availability (Giacomini et al, 2008).

Parameters according to nature and location of residues (contact with the soil)

Residue input

(Nicolardot et al, 2001; Justes et al., 2009)

Crop residues or organic wastes left as **mulch or buried** into the soil

/!\ *(measured remaining biomass may include microbial biomass!)*

Direct input (tec file) or automatic input **at harvest**

	ressuite	Qressuite	QNressuite
1	'whole_crop'	masec	QNplante-Nrac
2	'straw+roots'	mspaille	QNveget-Nrac
3	'stubble+roots'	mspaille*0.35	(QNveget-Nrac)*0.35
4	'prunings'	mabois	mabois *5
5	'stubble_of_residu_type_9+roots'	mspaille	QNveget-Nrac
6	'stubble_of_residu_type_10+roots'	mspaille	QNveget-Nrac

Contributions of **root** residues at harvest = f(root system description option; root length, specific parameter)

Humus mineralization (C and N)

Potential mineralization rate Vminh = K2*NHUM

$$K2 = K2pot * \frac{\sum_{z=1}^{profhum} FH*FTH}{profhum}$$

K2pot : 2 choices

Delphin, 1986 and Chaussod et al, 1986

$$k2pot = fmin1 * \frac{e^{-fmin2*argi}}{1. + fmin3 * calc}$$

Clivot et al, 2017

$$k2pot = gmin1 * \frac{e^{-gmin2*argi}}{(1. + gmin3 * calc)} * e^{-gmin4*(pH-gmin5)^2} * (0.8 * e^{-gmin6*(CsurNsol-gmin7)^2} + 0.2)$$

Normalized days

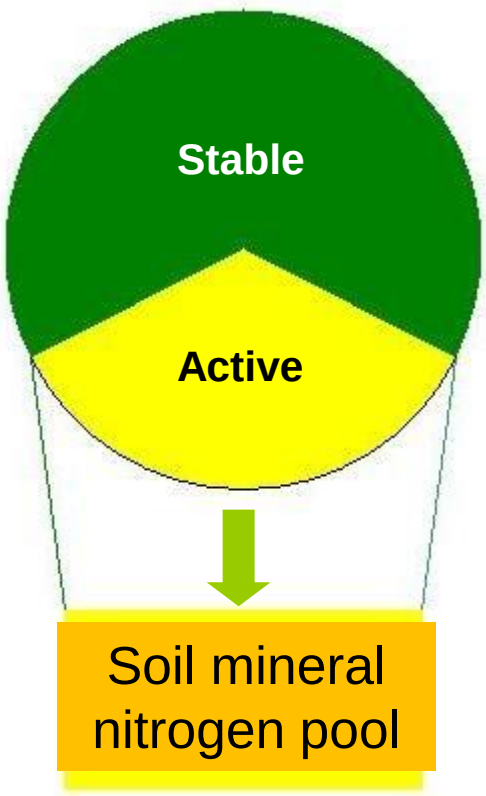
$$FTH = \exp\left(FTEM \ (T - T_{REF})\right)$$

$$FH = FHUM + (1 - FHUM)\frac{HUR - HUMIN}{HUCC - HUMIN}$$

$profhum$: thickness of mineralization layer ~ ploughed layer or high root density horizon

Humus mineralization (C and N)

Humus



stable/total = 0.65

(Andriulo et al., 1999)

Potential mineralization rate Vminh = K2*NHUM

$$NHUM_{*} = Norg * Da(1) * PROFHUM * (1 - FINERT)$$

$*active\ fraction$

$$CHUM_{*} = Corg * Da(1) * PROFHUM * (1 - FINERT)$$

Da : only on the first layer in the 9.0 version , will be corrected on the v10

$profhum$: thickness of mineralization layer ~ ploughed layer or high root density horizon

Gaseous N losses through volatilization, nitrification, denitrification

1) Empirical fertilizer efficiency approach of N losses:

immobilization – volatilization – denitrification

$$\mathbf{EFFN = 1 - \frac{NORGENG + NVOLENG + NDENENG}{ANIT}}$$

=> Efficiency = f(fertilizer typology, crop N uptake rate)

2) Mecanistic approach

Description and simulation of volatilization, nitrification and denitrification processes

Volatilization

- Purely **chemical process**
- Transform ammonium (NH4+) into **gaseous ammonia** (NH3)
- Only available for **organic manure**
- Tracks changes between 4 forms of ammonia compounds, one of them, NH3 in gaseous phase leads to volatilization.

Function of NH4+ concentration, temperature, evolving pH (high value = increase in volatilization), manure properties (Nmin, water content), soil tillage, soil porosity, wind speed...

(Génermont and Cellier, 1997)

Nitrification, denitrification and associated N2O emissions

- Options to activate (codenitrif, codedenit), but should be the default
- Use default values first
- Main uncertainty = denitrification potential
- Active over profhum or prodenit

N_2O module: nitrification

potential nitrification rate

regulation factors

nitrification rate

2

$$vnit = vnitpot . f_{NH4} . f_{pH} . f_T . f_{WFPS}$$

soil water content – aeration effect

NH_4 availability

pH effect

temperature effect

1

$$vnit = \boxed{fnx \ * NH4} . f_{pH} . f_T . f_{WFPS}$$

Option 2 is more satisfying but option 1 works better in practice because the absence of saturation compensates for unrealistic high NH4 concentration at the soil surface!

N_2O module: denitrification

f(Norg & CsurNsol)
or *vpotdenit*

potential denitrification rate

denitrification rate

regulation factors

$$vdenit = vdenitpot . f_{N03} . f_T . f_{WFPS}$$

NO_3 availability

temperature effect

soil water content – aeration effect

N_2O module: N_2O emissions

N_2O emissions correspond to a constant or variable proportion of nitrification and denitrification rates

water content – aeration effect

N_2O emissions associated to nitrification

$$N2Onit = z.vnit$$

nitrification rate

N_2O emissions associated to denitrification

$$N2Odenit = r.vdenit$$

denitrification rate

pH, water content-aeration and [NO_3] effects

N_2O module : options

	option	parameter name	default value	other value
NITRIFICATION	dependence on NH_4 for vnitpot	code_vnit	proportionality to NH_4 concentration	use of Michaelis-Menten function
	temperature function	code_tnit	Gaussian function	linear function
	N_2O ratio	code_rationit	variable with soil water saturation	constant
	hourly WFPS	code_hourly_wfps_nit	yes	no
DENITRIFICATION	source of denitrification potential	code_pdenit	estimated using Norg and CsurNsol	vpotdenit
	N_2O ratio	code_ratiodenit	variable with soil pH, WFPS and NO3 concentration	constant
	hourly WFPS	code_hourly_wfps_denit	yes	no

Thank you for your attention